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Department : CSE

Subject : Python

Batch : 3

Mini Project 1:Electric Bill Calculator using Python

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1. Project Title

Electric Bill Calculator using Python

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2. Problem Statement

Develop a Python program to calculate the electricity bill for multiple customers based on the number of units consumed.

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3. Objective

- Apply different tariff rates based on unit consumption.
- Demonstrate the use of loops to process multiple customers.
- Demonstrate the use of conditional statements for decision making.
- Perform basic arithmetic calculations without using arrays.
- Simulate a real-life electricity billing system in a simple manner.

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4. Tools Used

- Python 3
- VS Code / IDLE

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5. Description

This project is a simple Electricity Bill Calculator developed using Python. The program calculates the electricity bill for multiple customers based on the number of units consumed.

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6. Algorithm

- Start the program.
- Display the title Electricity Bill Calculator.
- Ask the user to enter the number of customers.
- For each customer, do the following:
  - Ask the user to enter the number of units consumed.
  - If the units are less than or equal to one hundred, calculate the bill at five rupees per unit.
  - If the units are between one hundred and two hundred, calculate the bill by charging five rupees per unit for the first one hundred units and seven rupees per unit for the remaining units.
  - If the units are more than two hundred, calculate the bill by charging five rupees per unit for the first one hundred units, seven rupees per unit for the next one hundred units, and ten rupees per unit for the remaining units.
  - Display the total bill amount for the customer.
- End the program.

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7. Program (Python Code)

```
print("Electricity Bill Calculator")


n = int(input("Enter number of customers: "))


for i in range(n):
    print("\nCustomer", i + 1)
    units = int(input("Enter units consumed: "))


    if units <= 100:
        bill = units * 5
```

```
    elif units <= 200:
        bill = (100 * 5) + ((units - 100) * 7)
    else:
        bill = (100 * 5) + (100 * 7) + ((units - 200) * 10)

    print("Total Bill Amount: ₹", bill)

print("\nAll bills generated successfully!")
```

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8. Sample Output

```
---------------------------- RESTART:
Electricity Bill Calculator
Enter number of customers: 3

Customer 1
Enter units consumed: 50
Total Bill Amount: ₹ 250

Customer 2
Enter units consumed: 150
Total Bill Amount: ₹ 850

Customer 3
Enter units consumed: 250
Total Bill Amount: ₹ 1700

All bills generated successfully!
```

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9. Result

The Electricity Bill Calculator program was successfully developed and executed using Python.

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10. Conclusion

The Electricity Bill Calculator program was successfully implemented using Python. The project demonstrates the effective use of looping and conditional statements to solve a real-life problem.